COL724/7524 Advanced Computer Networks

Assignment 1 – Part E
Routing Convergence

1 Objective

Part E studies routing convergence following a link failure and subsequent recovery. Continuous UDP traffic is generated from R1 to R5 so that changes in the selected route can be observed while the network is operating.

Two paths are available between R1 and R5:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R5$$

and

$$R1 \rightarrow R4 \rightarrow R5.$$

The R1–R4 link is failed at $t = 6$ s and restored at $t = 9$ s. The experiment measures packet delivery, delay, jitter, route changes, and routing convergence time.

2 Experimental Setup

The simulation uses a five-router topology with two paths between R1 and R5. Path A contains three hops,

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R5,$$

while Path B contains two hops,

$$R1 \rightarrow R4 \rightarrow R5.$$

Continuous UDP traffic is generated from R1 to R5 with a packet interval of 10 ms. The R1–R4 link is disabled at $t = 6$ s, forcing traffic to use the alternate path through R2 and R3. The link is restored at $t = 9$ s, after which traffic returns to the primary path.

FlowMonitor is used to collect packet transmission, reception, loss, delay, and jitter statistics.

The convergence time is calculated as

$$T_{\text{conv}} = T_{\text{first successful packet}} - T_{\text{event}}.$$

3 FlowMonitor Results

The FlowMonitor output identifies two flows corresponding to the two paths. Flow 1 corresponds to the primary R1–R4–R5 path, while Flow 2 corresponds to the alternate R1–R2–R3–R5 path. Flow 1 contains 1098 transmitted packets and 1097 received packets, while Flow 2 contains 366 transmitted and 366 received packets.

Table 1: Part E FlowMonitor results

Flow	**Observed Phase**	**Path**	**Tx**	**Rx**	**Lost**
Flow 1	Before and after recovery	R1–R4–R5	1098	1097	1
Flow 2	During failure	R1–R2–R3–R5	366	366	0

The alternate-path flow successfully delivers all 366 transmitted packets, resulting in zero packet loss during the failure interval. The primary-path flow has one lost packet over its aggregate measurement interval.

The average delay is calculated using

$$D_{\text{avg}} = \frac{\text{delaySum}}{\text{rxPackets}}.$$

The measured delay and jitter values are:

Table 2: Average delay and jitter

Flow	**Path**	**Average Delay (ms)**	**Jitter (ms)**
Flow 1	R1–R4–R5	6.87	0
Flow 2	R1–R2–R3–R5	10.30	0

The alternate path has a higher average delay because traffic travels through three routers instead of the shorter R1–R4–R5 path.

4 Route Change

The packet-level output confirms the route change during the failure interval. Before the failure, packets use the primary R1–R4–R5 path. When the R1–R4 link fails at $t = 6$ s, traffic switches to R1–R2–R3–R5. After the link is restored at $t = 9$ s, traffic returns to the primary path.

Table 3: Observed route changes

Phase	**Time**	**Selected Path**
Normal operation	$t < 6$ s	R1–R4–R5
Link failure	$6 \leq t < 9$ s	R1–R2–R3–R5
After recovery	$t \geq 9$ s	R1–R4–R5

The alternate-path flow is active approximately from 6.006 s to 8.996 s, which closely matches the period during which the R1–R4 link is unavailable. This provides evidence of successful route failover.

5 Routing Convergence

For the link failure at $t = 6$ s, the first successfully received packet on the alternate path occurs at $t = 6.01619$ s. Therefore,

$$T_{\text{conv,failure}} = 6.01619 - 6.00000 = \boxed{16.19\text{ ms}}.$$

For the link recovery at $t = 9$ s, the first successfully received packet using the restored R1–R4–R5 path occurs at $t = 9.01103$ s. Thus,

$$T_{\text{conv,recovery}} = 9.01103 - 9.00000 = \boxed{11.03\text{ ms}}.$$

Table 4: Part E convergence measurements

Event	**Event Time (s)**	**New Path**	**First Successful Packet (s)**	**Convergence**
Link failure	6.000	R1–R2–R3–R5	6.01619	16.19
Link recovery	9.000	R1–R4–R5	9.01103	11.03

The measured recovery convergence is approximately 5.16 ms faster than the convergence following the link failure.

6 Discussion

The experiment demonstrates that the network can maintain connectivity when the primary R1–R4 link fails. Continuous traffic allows the route transition to be observed directly. During the failure interval, traffic is transferred to the longer R1–R2–R3–R5 path.

The alternate path introduces higher delay, with an average FlowMonitor delay of approximately 10.30 ms compared with 6.87 ms for the primary path. Nevertheless, all packets belonging to the alternate-path flow are successfully delivered.

After the R1–R4 link is restored at $t = 9$ s, traffic returns to the primary path. The measured convergence times are 16.19 ms for failure and 11.03 ms for recovery.

7 Conclusion

Part E demonstrates routing convergence under continuous traffic. When the R1–R4 link fails at $t = 6$ s, traffic successfully switches to R1–R2–R3–R5. When the link is restored at $t = 9$ s, traffic returns to R1–R4–R5.

The measured convergence times are 16.19 ms after failure and 11.03 ms after recovery. FlowMonitor confirms successful delivery over the alternate path, demonstrating that the network remains connected despite the link failure.